

CLIMATE CHANGE IMPACT ON ENERGY DEMAND

Analyzing Climate, Economic and Energy Indicators Across 192 Countries (2001–2022)

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192
Countries

2001–2022
Time Period

3,300
Records

Best Model
Random Forest

$R^2 = 0.83$
Accuracy

1. Research Problem & Objective

Problem

Climate change is altering electricity demand patterns worldwide.

Rising temperatures increase cooling-related energy consumption.

Energy systems require improved forecasting and planning strategies.

Objective

Analyze how climate, economic, and demographic indicators influence electricity demand across 192 countries.

2. Research Questions & Hypotheses

Research Questions

- RQ1: How does climate change influence electricity demand across countries?
- RQ2: What relationships exist among temperature, CO₂ emissions, GDP, and electricity demand?
- RQ3: Can climate and economic indicators predict future electricity demand?

Hypotheses

- H1: Temperature ↑ → Electricity Demand ↑
- H2: CO₂ ↑ → Electricity Demand ↑
- H3: GDP & Renewable Energy significantly influence electricity demand

3. Dataset & Preprocessing

Dataset Overview

- Countries: 192
- Period: 2001–2022
- Records: 3,300
- Variables: 15+ Climate, Economic & Energy Indicators
- Target: Electricity Demand Per Capita

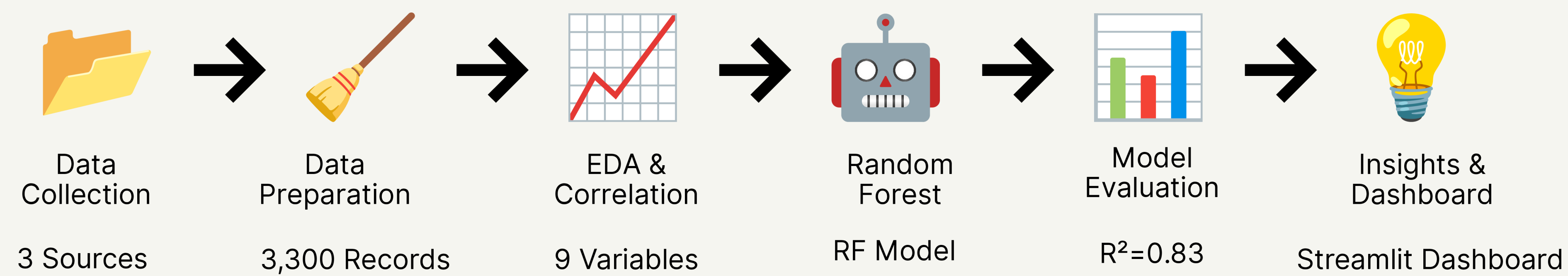
Data Sources

- FAOSTAT - Temperature Change
- OWID - Energy Data, Renewable Energy
- OWID - CO₂ Emissions, GDP & Population

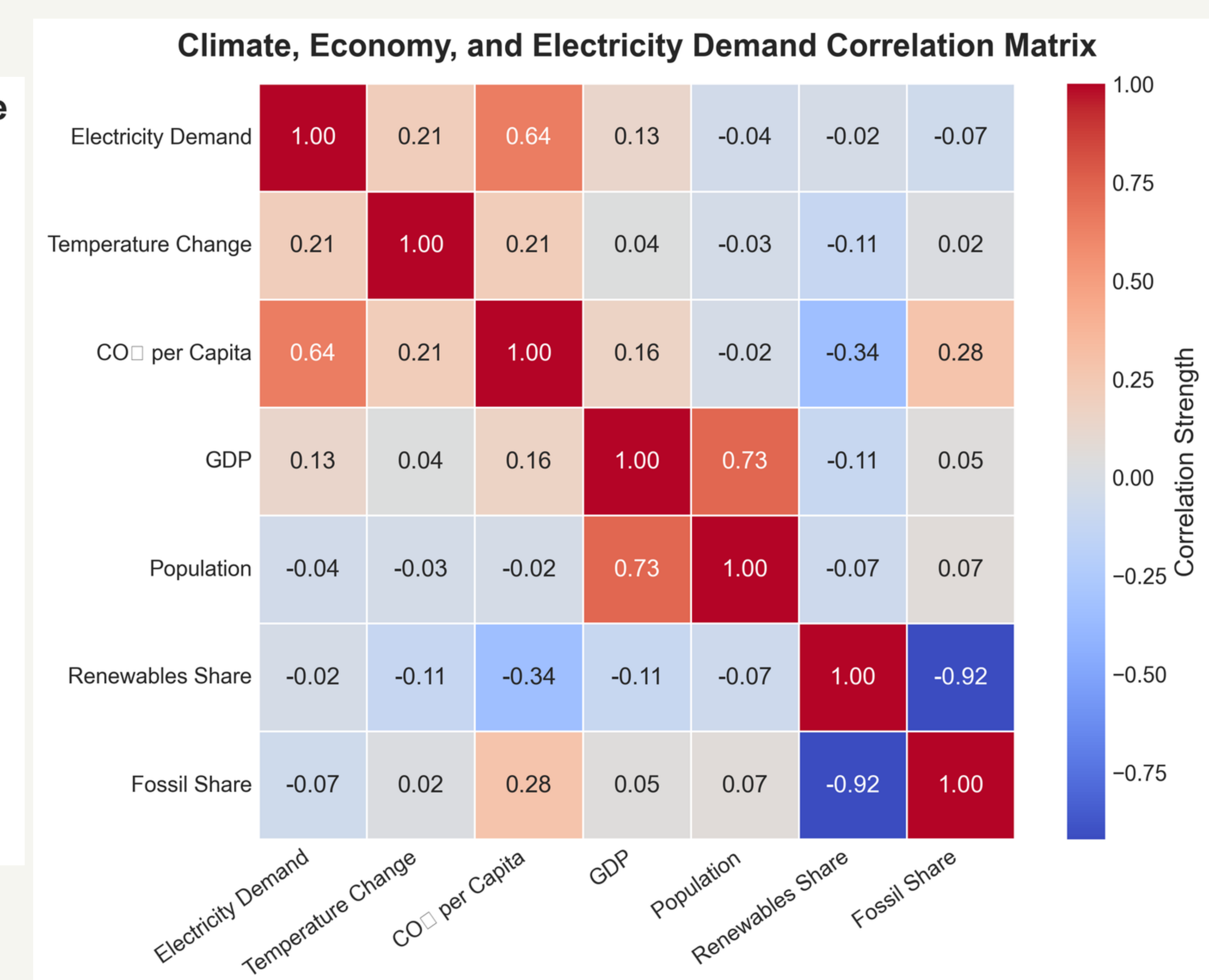
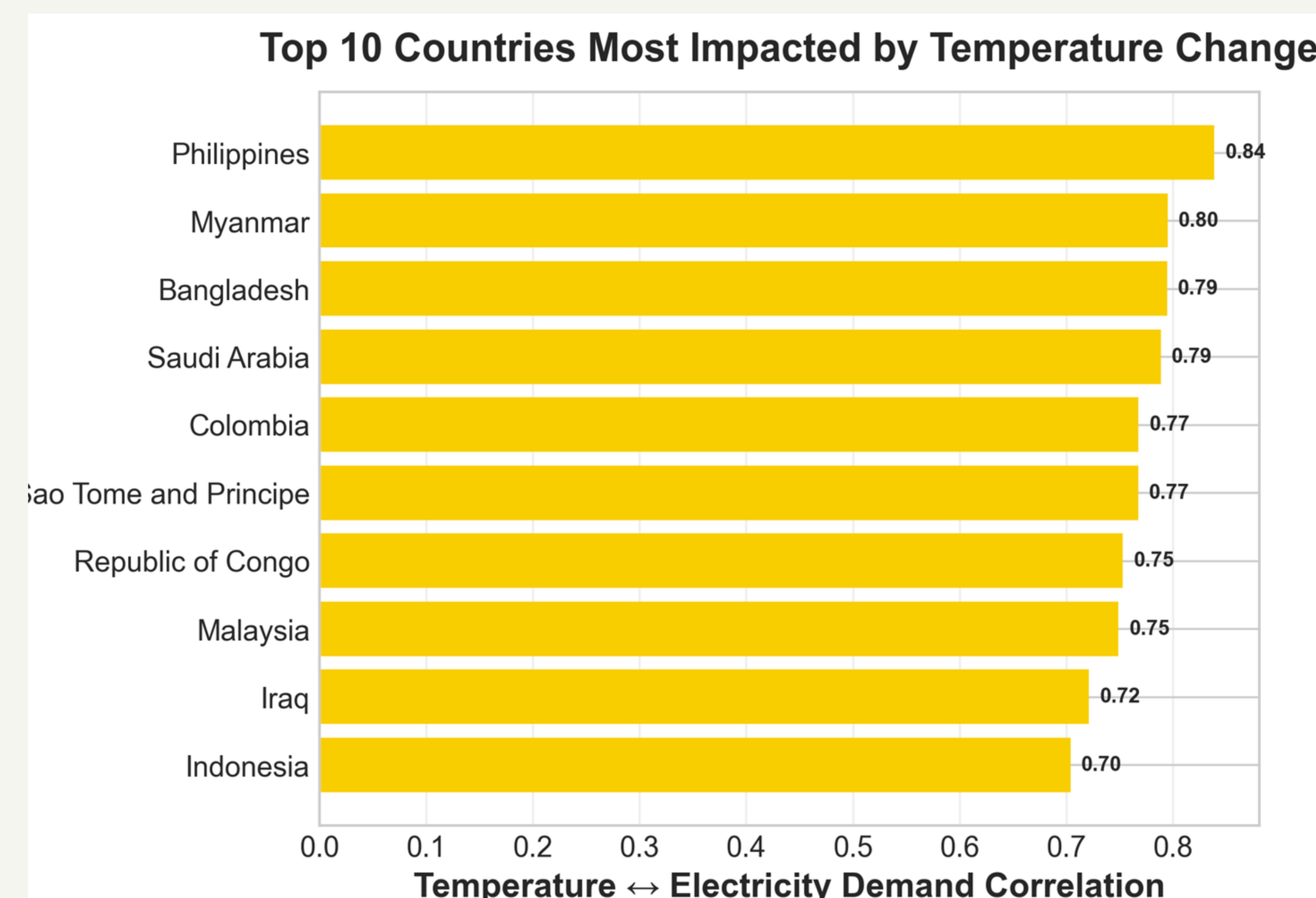
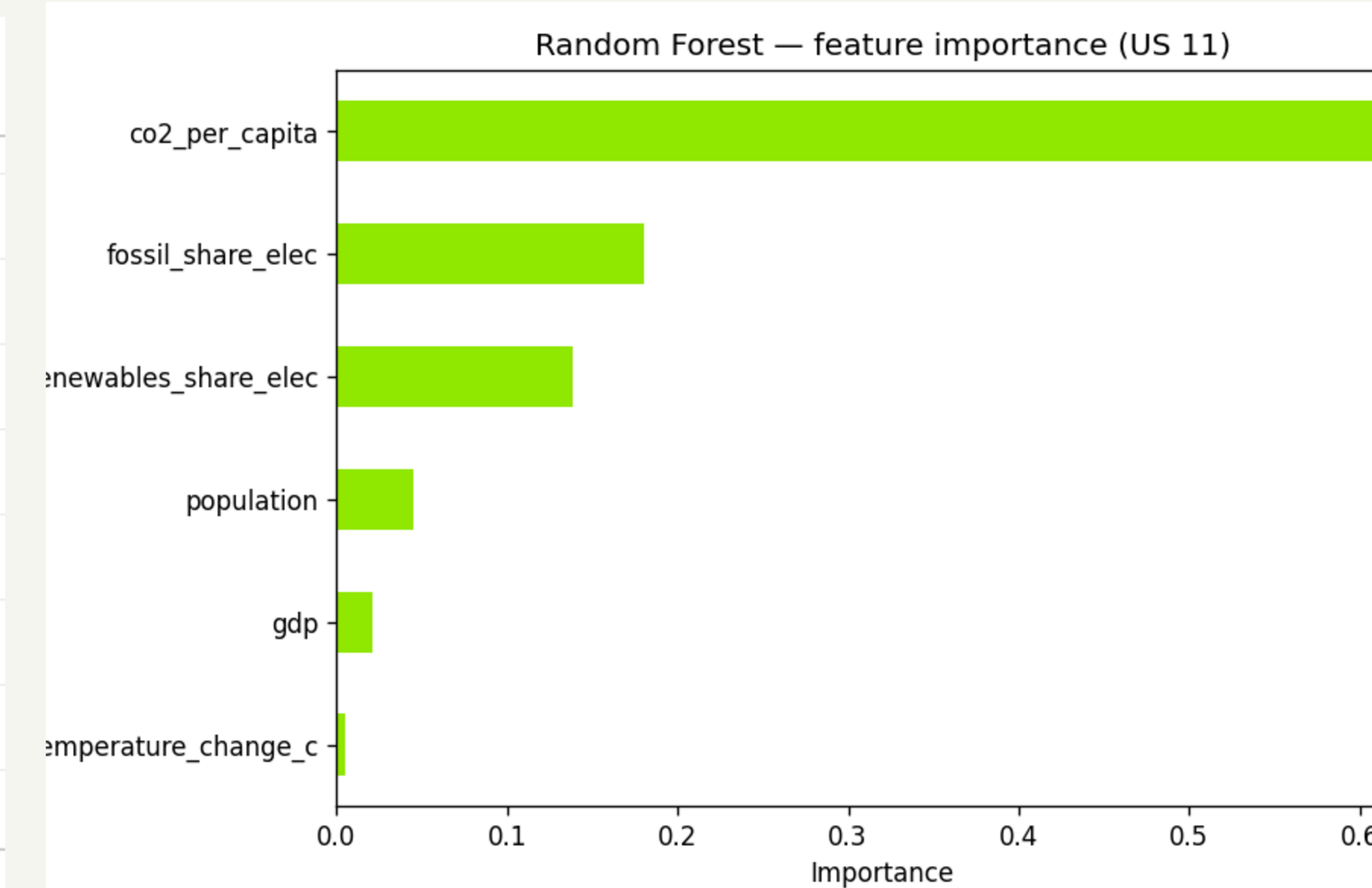
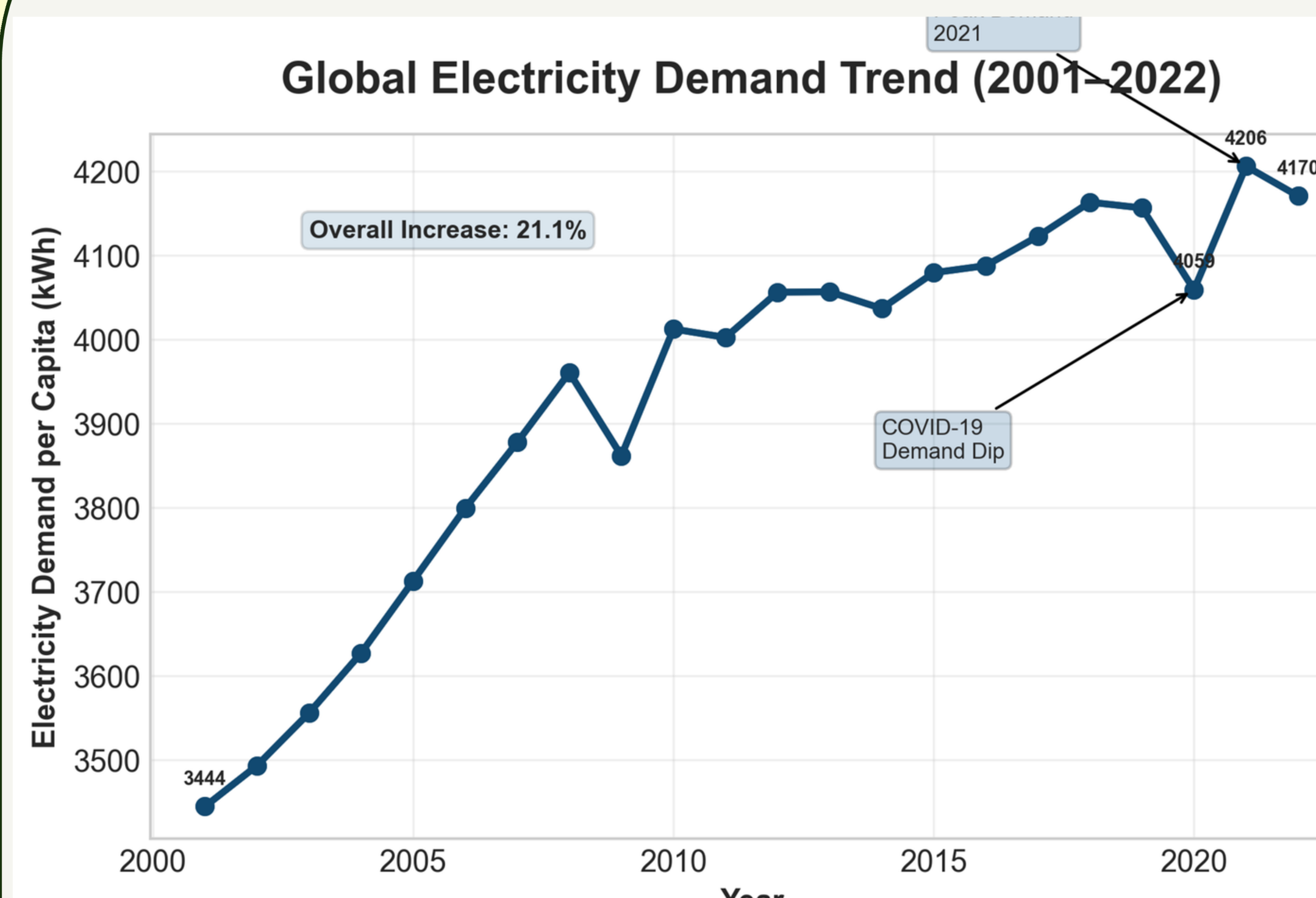
Processing Pipeline

Data Collection → Integration → Cleaning → Feature Engineering → Quality Checks

4. Analytical Methodology



5. Data Visualizations



6. Key Findings & Interpretations

Key Findings

- Random Forest: $R^2 = 0.83$ (Best Model)
- CO₂ & Energy Factors: Strongest Predictors of electricity demand
- Temperature Change: Significant Impact on Demand
- Climate + Economic Variables: Improve Prediction Accuracy
- Cross-Country Variation: Demand Patterns Differ by Region

7. Strategic Implications

Energy Policy

Climate-informed long-term planning

Utility Planning

Improve demand forecasting & grid reliability

Sustainability

Support renewable energy expansion

Climate Adaptation

Develop region-specific energy strategies

8. Limitations

Limitations

- Annual country-level data only
- Extreme weather events not modeled
- Future climate uncertainty remains

9. Future Research

Future Research

- Monthly/Daily Demand Forecasting
- Deep Learning Models
- Regional & City-Level Analysis
- Climate Scenario Evaluation

10. Explore Project



Scan to View
Interactive streamlit
Dashboard